

Automated Formulation & Wood Stain Color Prediction

Project Overview

A camera-based wood-stain color prediction system is very feasible. The key is to treat it as a **controlled color-measurement + prediction problem**, not simply “take a picture and have AI guess the stain.”

A practical system would look like this:

Camera → Color calibration → Wood identification → Stain prediction model → Formula recommendation → Predicted finished-color image

1. Start with controlled photography

Use a fixed camera rather than a phone held by hand.

You want:

- Fixed camera distance and angle
- Fixed exposure
- Fixed white balance
- Consistent lighting
- Neutral background
- A color-calibration target in every image
- Ideally RAW images rather than JPEG

For example:

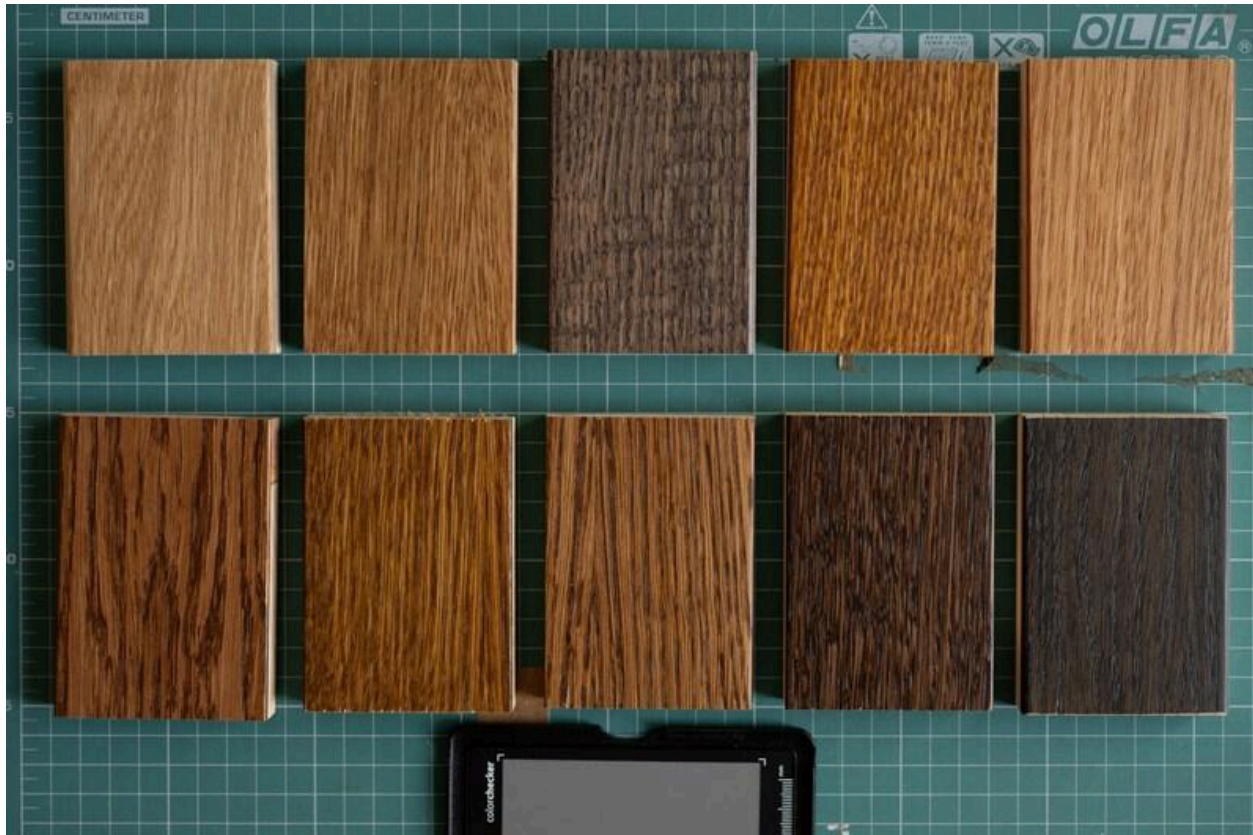
Color Chart Reference





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A **ColorChecker-type reference** in the image allows the software to correct differences between cameras and lighting.

2. Measure the unfinished wood

The software should first determine the characteristics of the wood itself.

Useful inputs include:

- Wood species
- Natural wood color
- Grain direction
- Porosity
- Growth-ring structure
- Existing finish, if any
- Sanding condition
- Moisture content, if available

This matters because the same stain formula can look dramatically different on maple, oak, cherry, walnut, etc.

3. Convert camera RGB into a useful color space

Don't build the prediction model directly around ordinary RGB values.

Convert the calibrated image into something like:

CIE L*a*b*

or preferably a perceptually useful space such as:

OKLab

Then your system can quantify color differences using values such as ΔE .

For example:

None

Unstained wood

$L^* = 72.4$

$a^* = 8.1$

$b^* = 24.6$

Target stain

$L^* = 54.7$

$a^* = 12.8$

$b^* = 28.1$

The prediction system then learns how a particular stain formulation changes the wood's color.

4. Build a training database

This is probably the most important part of the project.

You need physical samples where you know:

Wood + stain formula + application conditions → measured final color

For example:

Spec ies	Base Wood	Dye A	Dye B	Pigm ent	Stain	Final L*	Final a*	Final b*
Mapl e	Natural	2%	0.5 %	1%	Formula 101	61.2	11.4	29.7
Mapl e	Natural	3%	0.5 %	1%	Formula 102	58.7	12.1	27.3
Oak	Natural	2%	0.5 %	1%	Formula 101	55.1	10.7	25.9

You could eventually have **thousands of samples**.

5. Use a spectrophotometer if possible

This would make your system substantially better.

The camera provides the visual information, while a spectrophotometer provides the ground-truth color measurement.

You could create:

Camera image → predicted Lab

and compare that against:

Spectrophotometer → actual Lab

The difference becomes your calibration/training target.

6. Train the prediction model

There are actually two different AI problems you could solve.

A. Predict the result of a known formula

Input:

None

Camera image
Wood species
Stain formula
Concentration
Application parameters

Output:

None

Predicted L*a*b*
Predicted RGB image



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ΔE confidence

This is the easier and more useful first version.

B. Reverse-engineer a stain formula

This is much more interesting.

Input:

None

Photo of target wood/stain

Output:

None

Recommended stain formula

For example:

None

Target:

L* 52.8

a* 13.4

b* 25.7

Recommended starting formula:

Dye Brown	2.10%
Dye Red	0.42%

Dye Yellow	0.18%
Black pigment	0.06%
Vehicle	Balance

The system could then predict:

None

Predicted:

L* 53.1

a* 13.2

b* 25.9

Estimated ΔE : 0.38

7. I would use a hybrid AI system

Rather than relying entirely on a neural network, I'd build it as:

Computer vision + color science + machine learning + formulation optimizer

Something like:

None

CAMERA



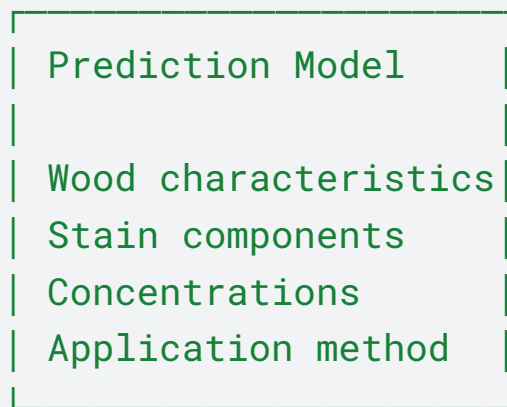
Color calibration



Wood/sample segmentation

↓
Extract calibrated color

↓
OKLab / Lab



↓
Color prediction

↓
 ΔE error calculation

↓
Formula optimization

↓
Recommended formulation

8. Add a "virtual stain" preview

This could be one of the most valuable features.

The user photographs an unfinished cabinet door.

The software displays:



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Original

→ **Predicted result with Wineberry**

→ **Predicted result with Espresso**

→ **Predicted result with Custom Formula 27**

That would make the system useful for customers, finishers, and formulators.

9. Application conditions need to be recorded

Don't overlook this.

The same formula can produce different colors depending on:

- Spray vs. wipe
- Wet film thickness
- Number of coats
- Flash time
- Drying conditions
- Wood sanding grit
- Wood species
- Sealer
- Topcoat
- Stain concentration

I'd therefore make the database record something like:

None

WOOD

Species: Maple

Sanding: 180 grit

STAIN

Formula: WB-042

Concentration: 8%

APPLICATION

Method: Spray

Gun: Kremlin

Pressure: XX

Wet film: XX

Flash: XX minutes

TOPCOAT

Product: Water-based urethane

Sheens: 20%

COLOR

L*: XX

a*: XX

b*: XX

ΔE : XX

That becomes extremely valuable training data.

10. Recommended technology stack

For a first prototype, I would use:

Hardware

- Industrial/DSLR camera
- Controlled LED lighting enclosure
- Color calibration target
- Spectrophotometer
- Computer

Software

- Python
- OpenCV
- NumPy
- scikit-learn
- PyTorch
- PostgreSQL or SQLite
- FastAPI
- React for the interface

You don't need an expensive AI system initially.

A gradient-boosting model or neural network trained on calibrated Lab data could produce a surprisingly good prototype.

11. The best way to start

I would **not** start by trying to predict every possible wood stain.

Start with perhaps:

3–5 wood species × 20–30 stain formulas × several concentrations × controlled application conditions.

That gives you a manageable experimental dataset.

Then test whether the camera can predict the final color within an acceptable ΔE tolerance.



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Once that works, expand the database.

An especially interesting version for AWFI

Given your water-based finishing work, you could build this specifically around **water-based wood finishing**.

The finished system could have a screen like:

TARGET COLOR

 *Take photo*

WOOD

Oak

DESIRED COLOR

Upload target / photograph sample

SYSTEM RECOMMENDATION

None

Formula: AWFI WB-042

Brown Dye	1.82%
Red Dye	0.31%
Yellow Dye	0.14%
Black Pigment	0.04%

Predicted ΔE : 0.7



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Confidence: 94%

Then show a **virtual rendering of the cabinet door with the predicted stain.**

That could evolve into a real **AI-assisted stain formulation and color-matching platform**, rather than simply a color-identification app.

[Open Source Spectrophotometer](#)